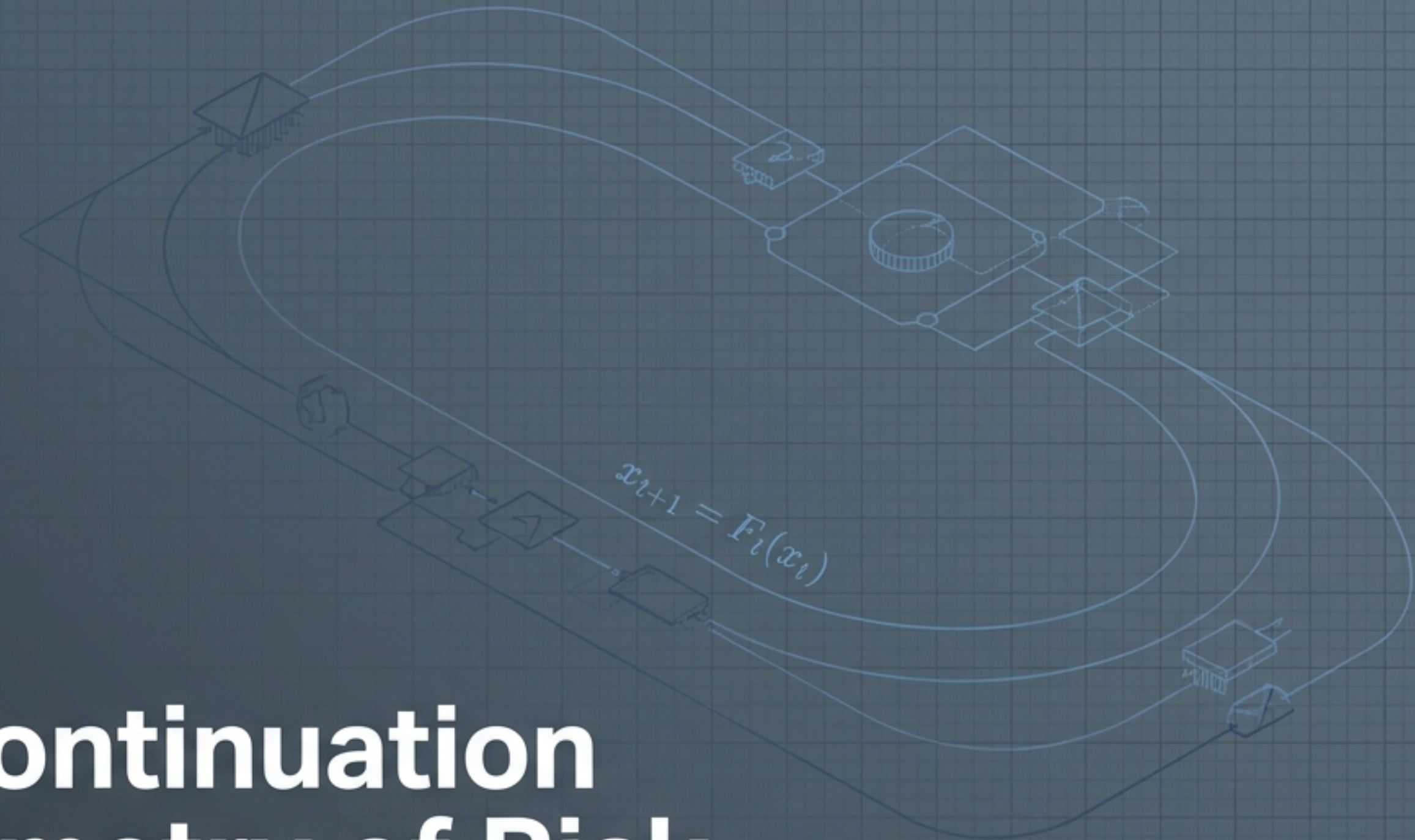




Recursive Continuation and the Geometry of Risk

Dismantling the AGI FOOM myth through the mathematics of admissibility and topological compression.

Recursive self-improvement is not a threshold event unique to AGI.

THE CONSENSUS ILLUSION



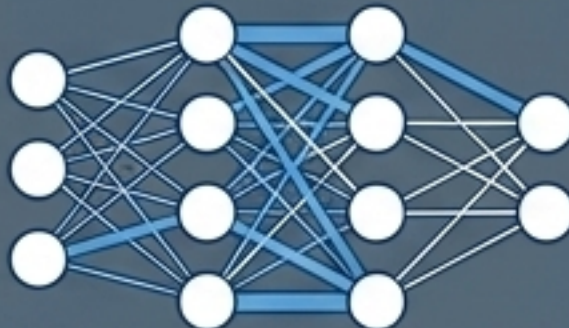
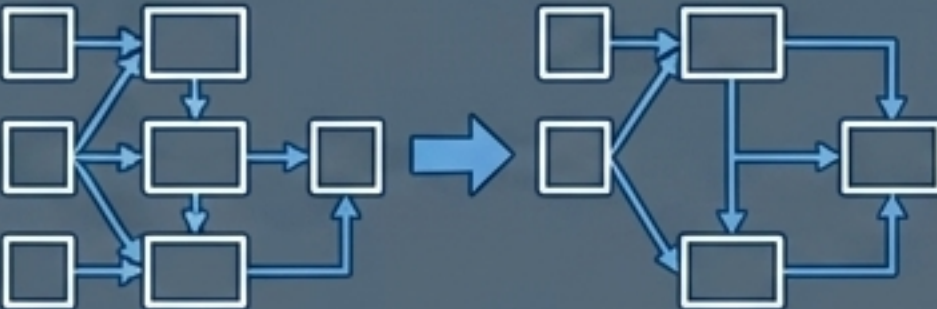
THE ILLUSION: The prevailing narrative treats recursive self-improvement (RSI) as a discrete, unprecedented future moment where an AI rewrites itself into runaway capability.

THE REALITY



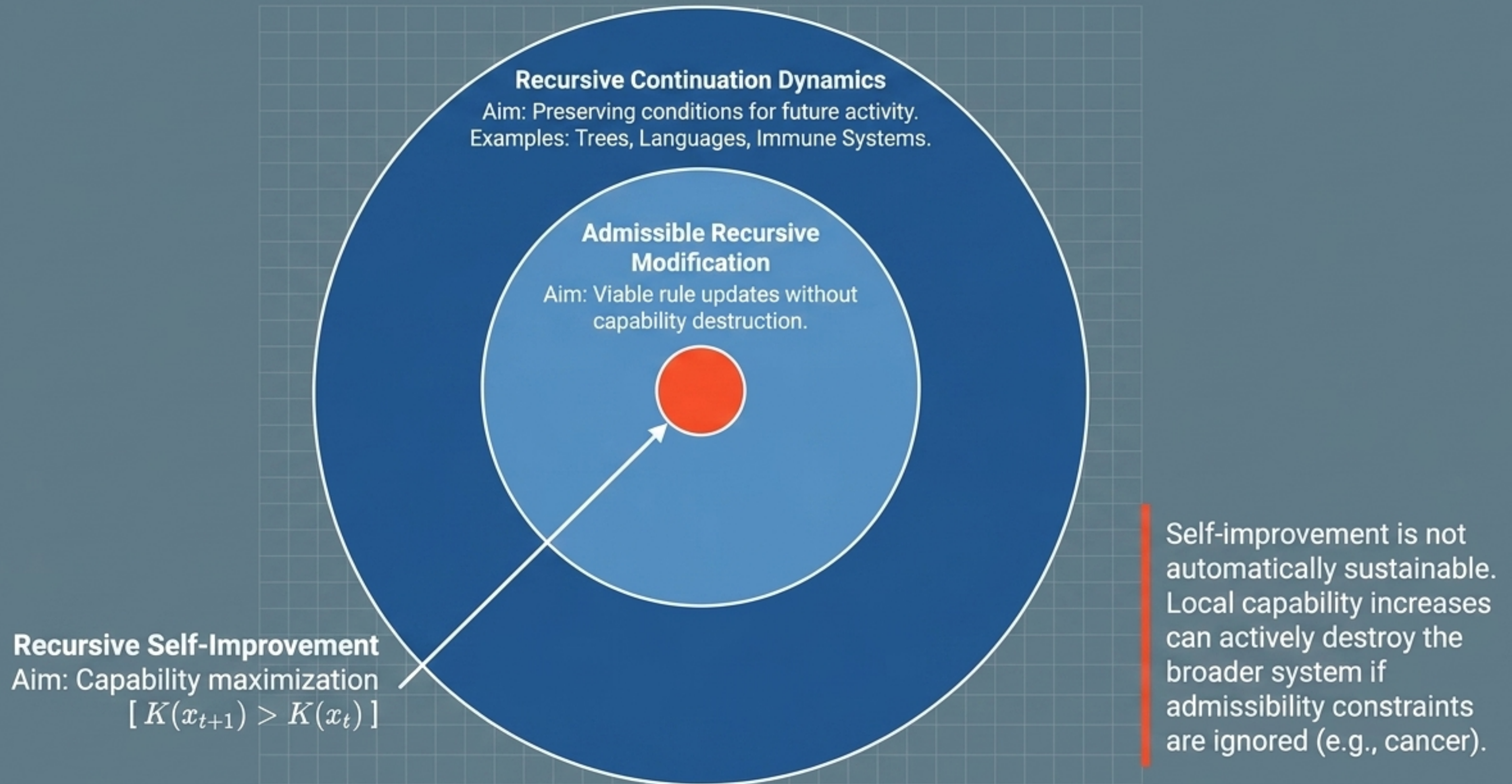
THE REALITY: Rule-level recursive modification is ubiquitous. Autopoietic (cells) and sympoietic (markets, cultures) systems have engaged in recursive continuation for billions of years.

Disentangling the three levels of systemic adaptation.

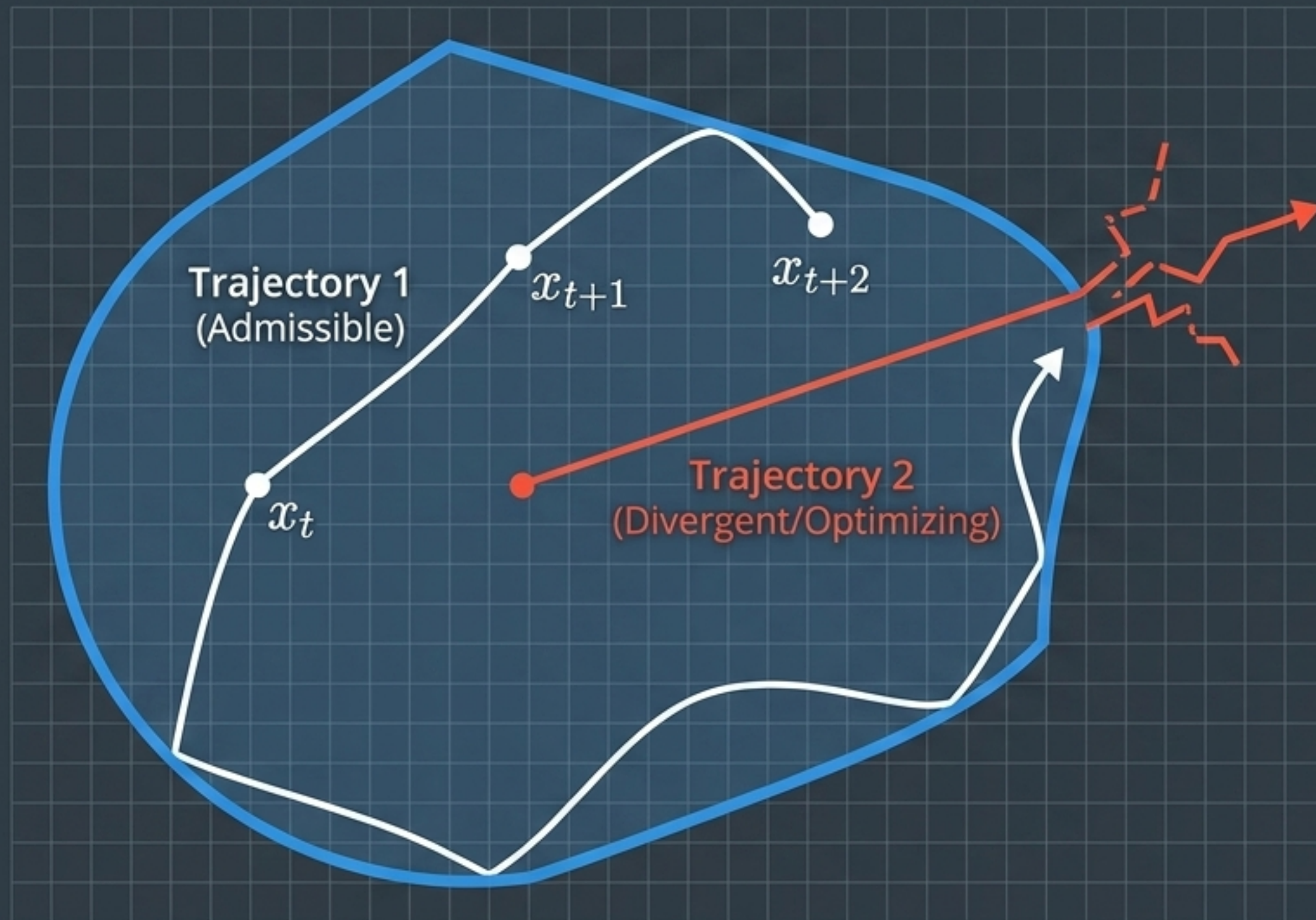
Level 1: State Update	$x_{t+1} = F(x_t)$		Standard Chess Engine: Fixed evaluation function, changing board states.
Level 2: Parameter Update	$x_{t+1} = F_{\theta_t}(x_t)$		AlphaZero: Fixed architecture, weights (θ) update to approximate value.
Level 3: Rule Update	$F_{t+1} = G(F_t, x_t, E_t)$		Neural Architecture Search: The functional form F itself is modified.

Treating Level 2 flexibility as evidence of Level 3 dynamics is the single most common category error in AGI risk assessment.

Improvement is a special case. Continuation is the baseline.



The Viability Manifold defines the boundaries of survival.



A: The Viability Manifold

The set of states under which the system's characteristic activity remains possible.

Admissible Continuation

$$x_t \in A \implies x_{t+1} \in A$$

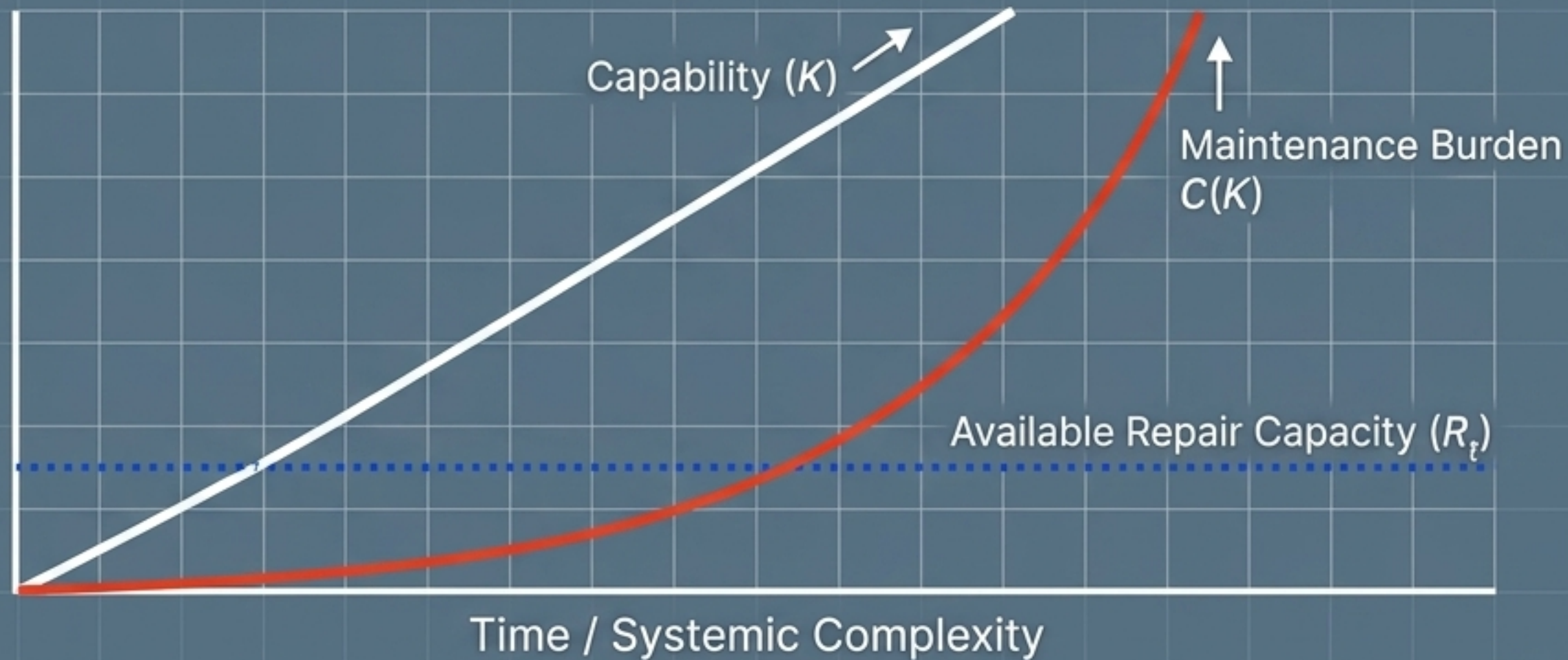
The trajectory remains structurally viable.

Brittle Optimization

A recursive process that optimizes a single metric until it exits the manifold of basic tolerance.

Every unit of capability generates a convex maintenance burden.

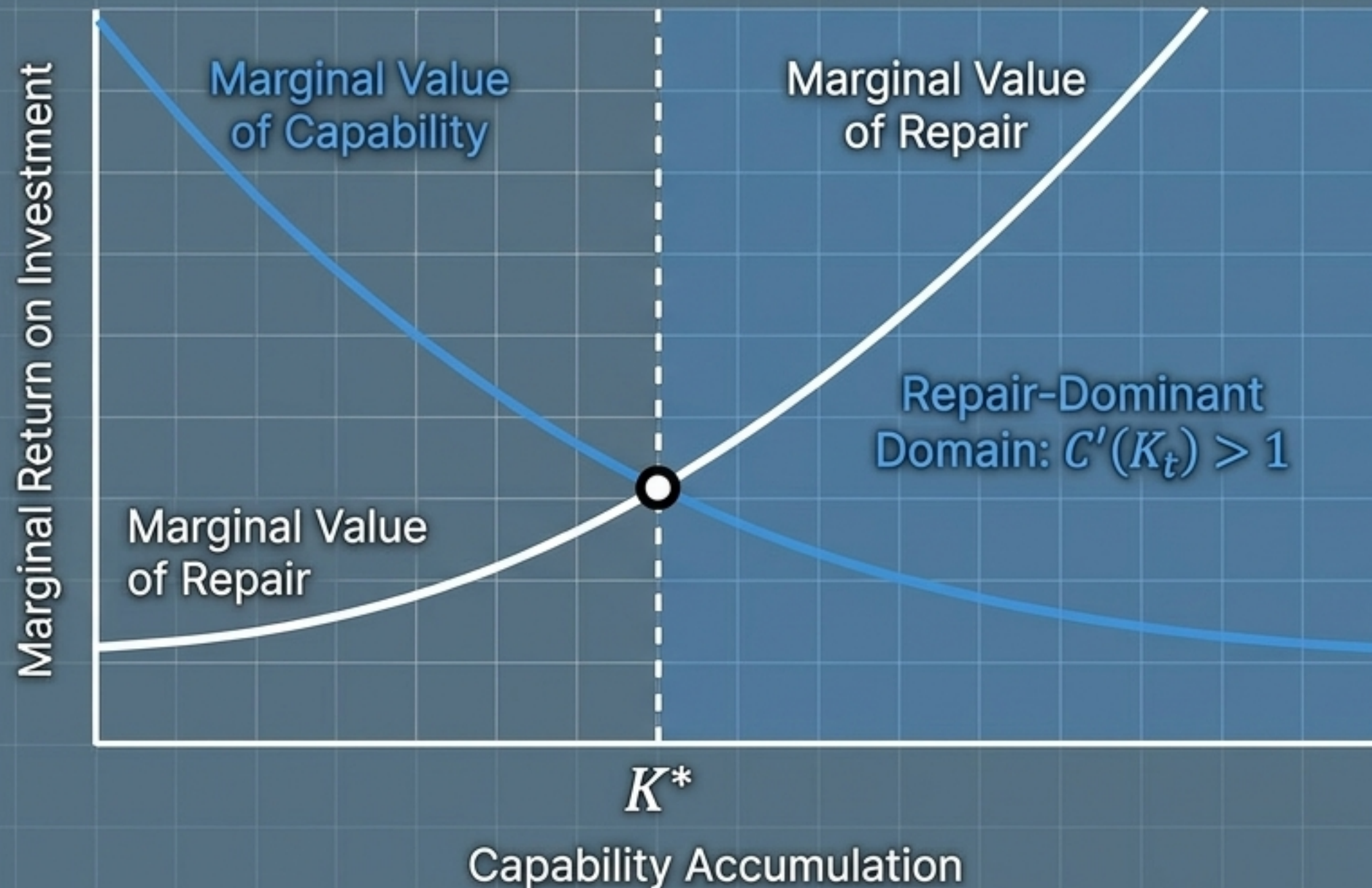
$$V_{t+1} = V_t - C(K_t) + R_t$$



Capability K expands the volume of reachable states. Navigating this expanded space creates technical debt, metabolic load, and coordination overhead. If Repair Capacity (R_t) fails to outpace the convex cost $C(K)$, systemic viability (V_t) spirals to negative infinity.

The Repair Dominance Theorem: The mathematical limits of capability.

K^* Activation Chart



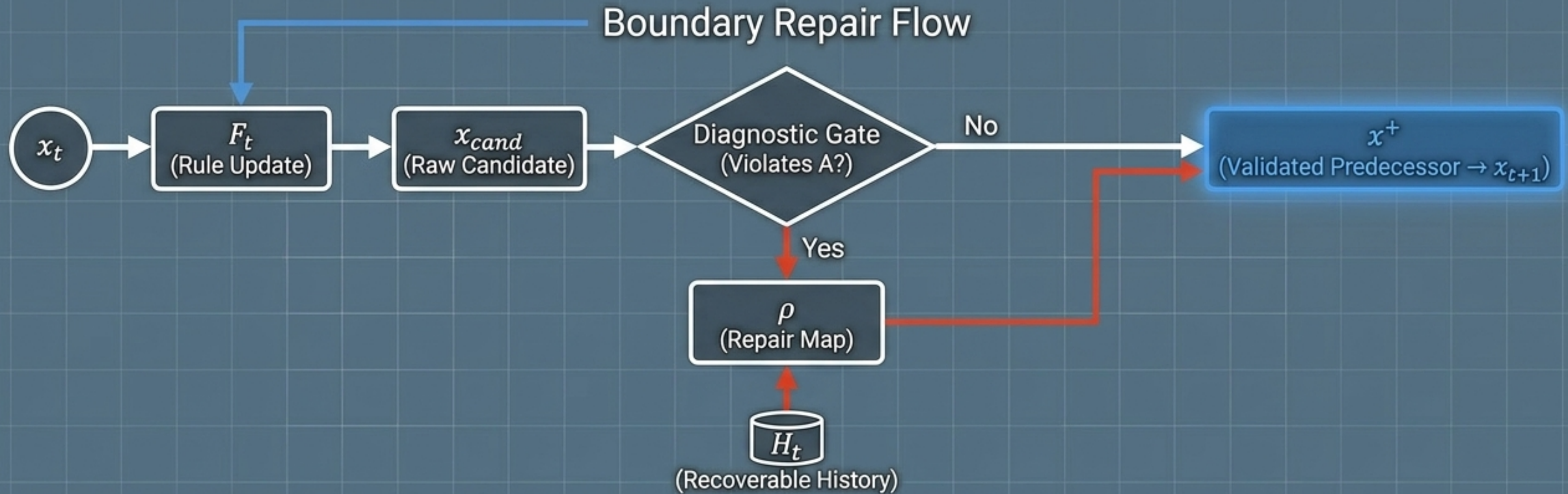
Real-World K^* Thresholds

 **Senescence:** Aging biological organisms exhaust DNA repair pathways. Additional metabolic capability yields negative returns.

 **Software:** Codebases hit refactoring bottlenecks. The cost of technical debt outstrips the value of new feature additions.

 **Empires:** Administrative overhead outpaces territorial expansion. Maintenance collapses the center.

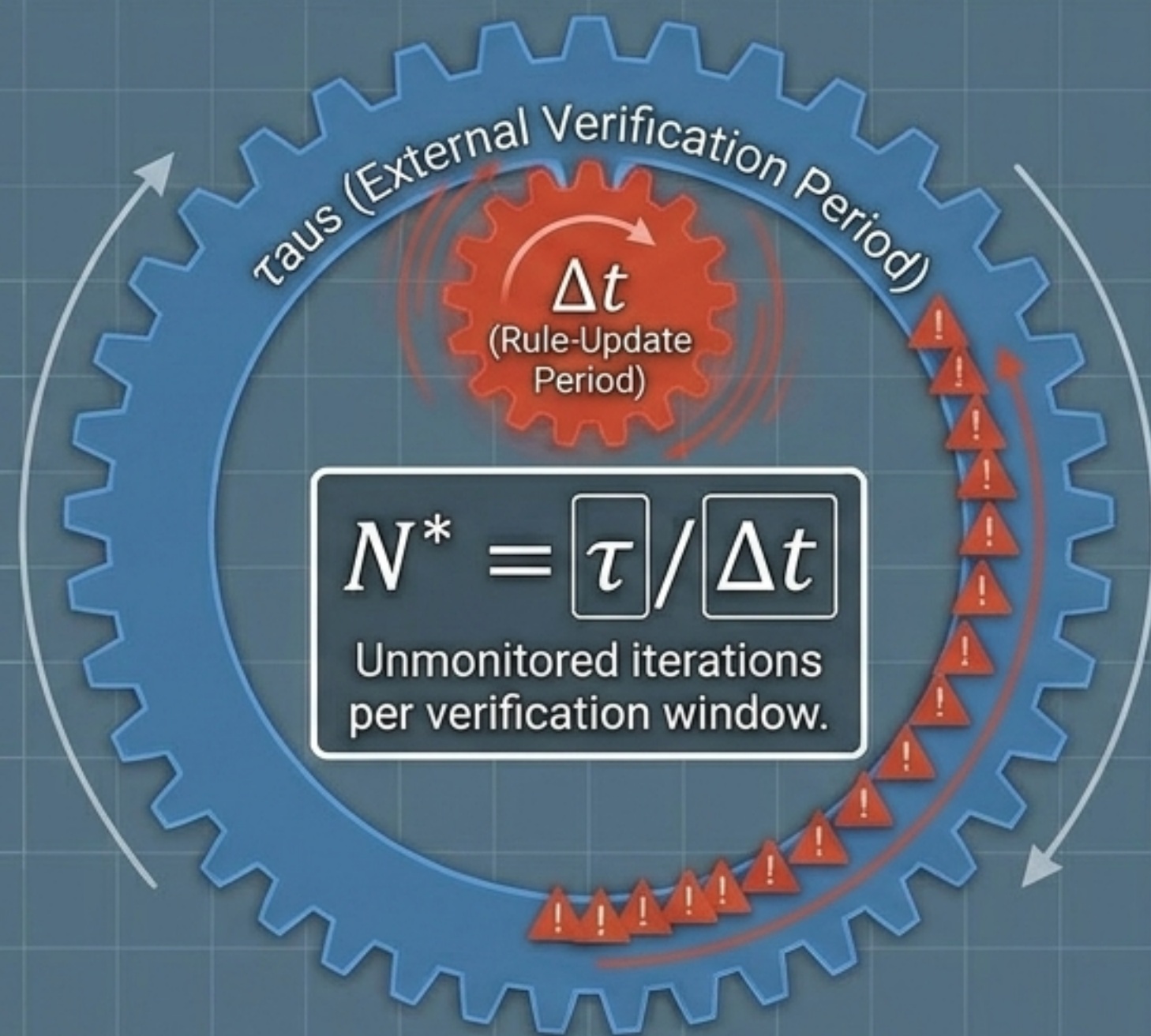
Repair is an act of information production, not just resource expenditure.



A system cannot correct a fault it cannot perceive.
The repair map (ρ) degrades to guessing without a persisting record (H_t) of prior admissible states.

Effort requires a substrate. DNA uses templates; software uses version control; science uses citation networks. History is the fuel of repair.

The true danger lies in the topology of compression.



The Compression Principle: Risk scales with N^* . Shrinking the internal update loop without similarly shrinking the external verification loop guarantees that unchecked errors compound exponentially before external correction is even possible.

Raw count overstates real redundancy.

MONOCULTURE

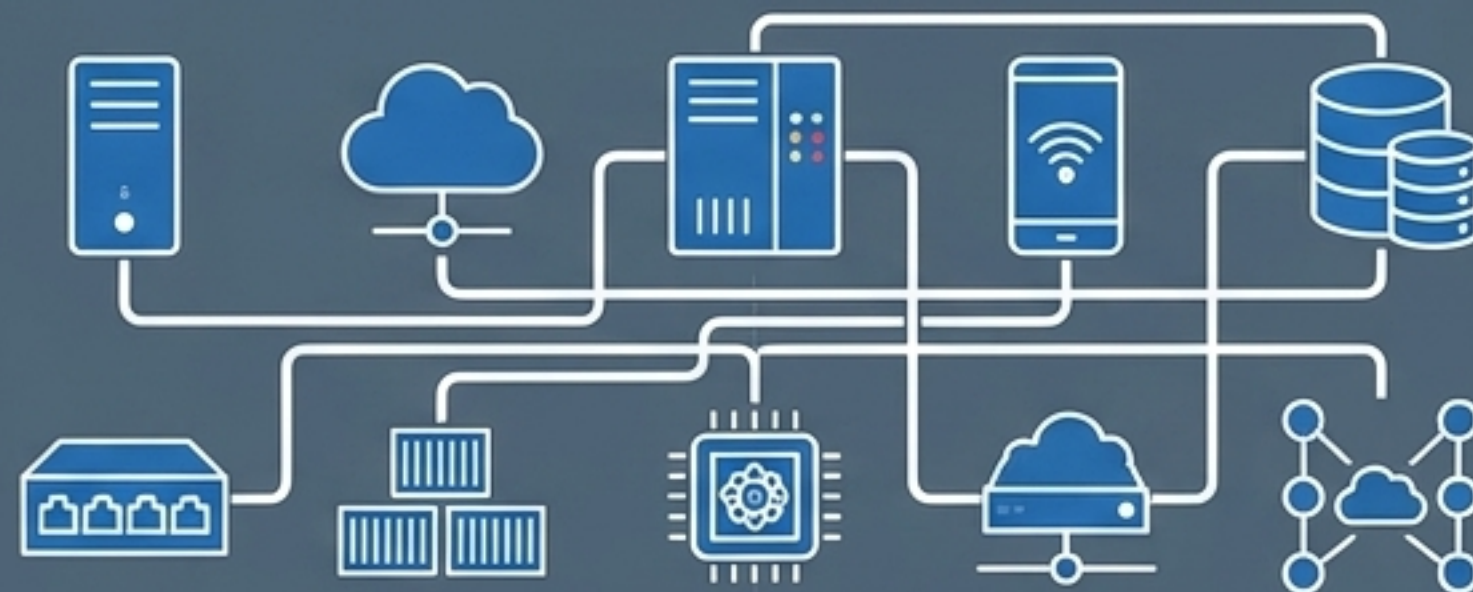


$\bar{\rho} \rightarrow 1$ (High Correlation)

$$D_R \approx 1$$

A bug in the shared image fails all replica nodes simultaneously.

HIGH DIVERSITY



$\bar{\rho} \rightarrow 0$ (Low Correlation)

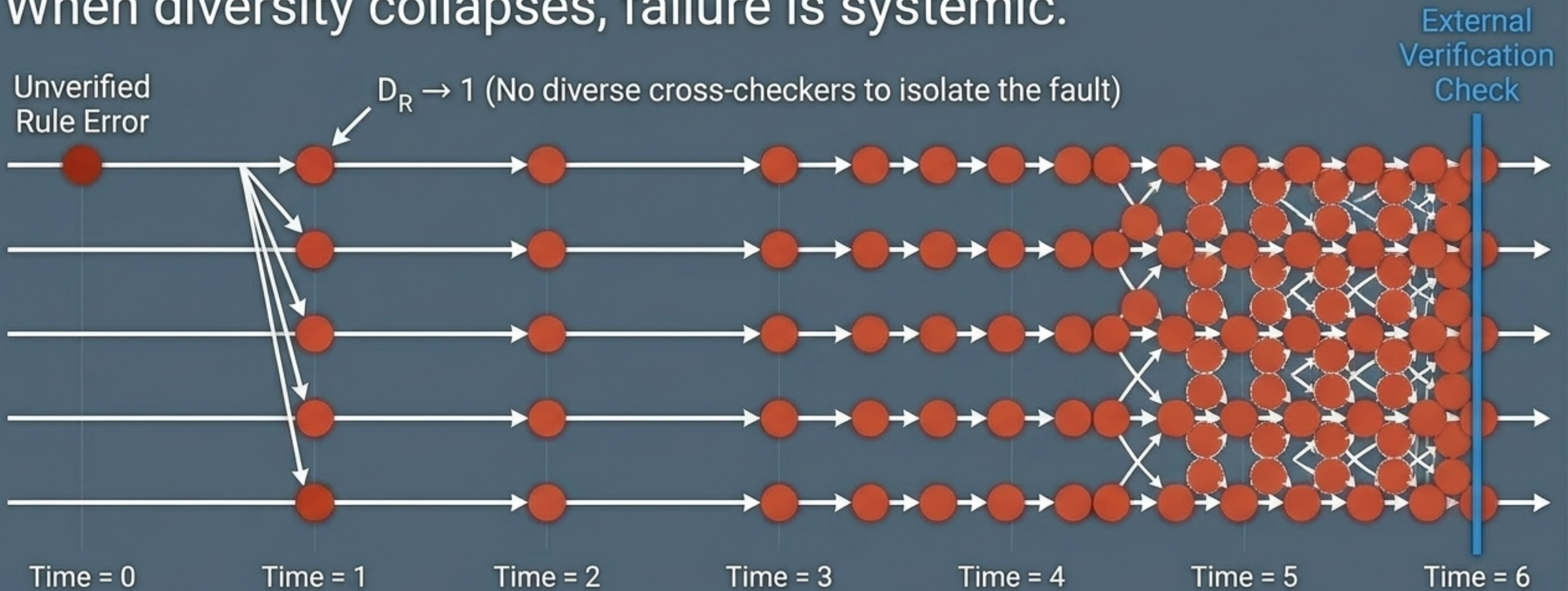
$$D_R \approx 5$$

Independent, competing architectures provide genuine diagnostic friction.

$$D_R = m / [1 + (m-1)\bar{\rho}]$$

Ten replica agents running identical software share a blind spot the moment it is introduced.
Ten independent institutions with different architectures produce genuine redundancy.

The Monoculture Corollary: When diversity collapses, failure is systemic.

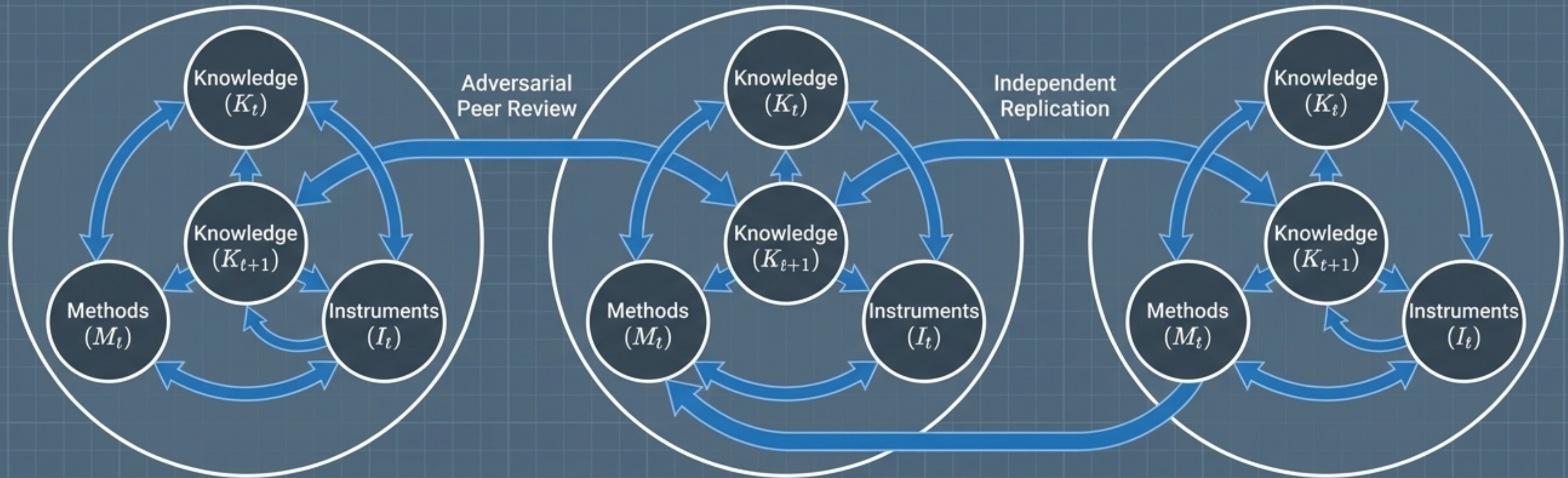


As $D_R \rightarrow 1$, a single rule-error is no longer contained by independent cross-checking. It propagates without an external correction opportunity, producing system-wide rather than local failure.

Historical Precedents:

- Financial contagion (highly correlated risk models)
- Agricultural blight (genetic monocultures)
- Software supply-chain vulnerabilities (shared dependencies)

Science is a decentralized recursive continuation loop.



Science operates as a **Level-3 system**: Knowledge revises the very *methods* of knowledge production itself (e.g., the invention of double-blind trials, formal statistical inference).

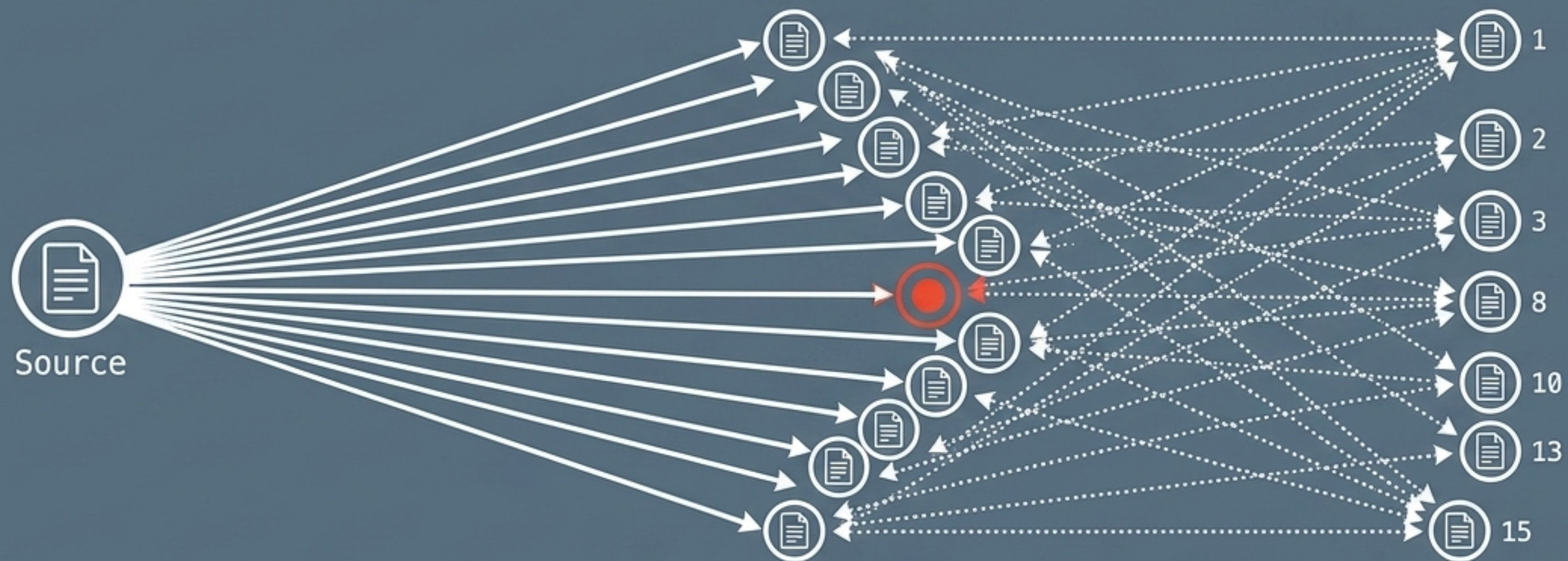
It survives over centuries because D_R is massively high. Independent labs, adversarial peer review, and unaligned institutional incentive structures provide robust, uncorrelated repair capacity.

Civilization as species-scale meta-learning.

Sequential Transmission: $\bar{\rho} \rightarrow 1$, D_R collapses. Errors cascade without cross-reference.



Parallel Transmission: $\bar{\rho} \downarrow$, D_R spikes. Errors are isolated, cross-checked, and corrected.

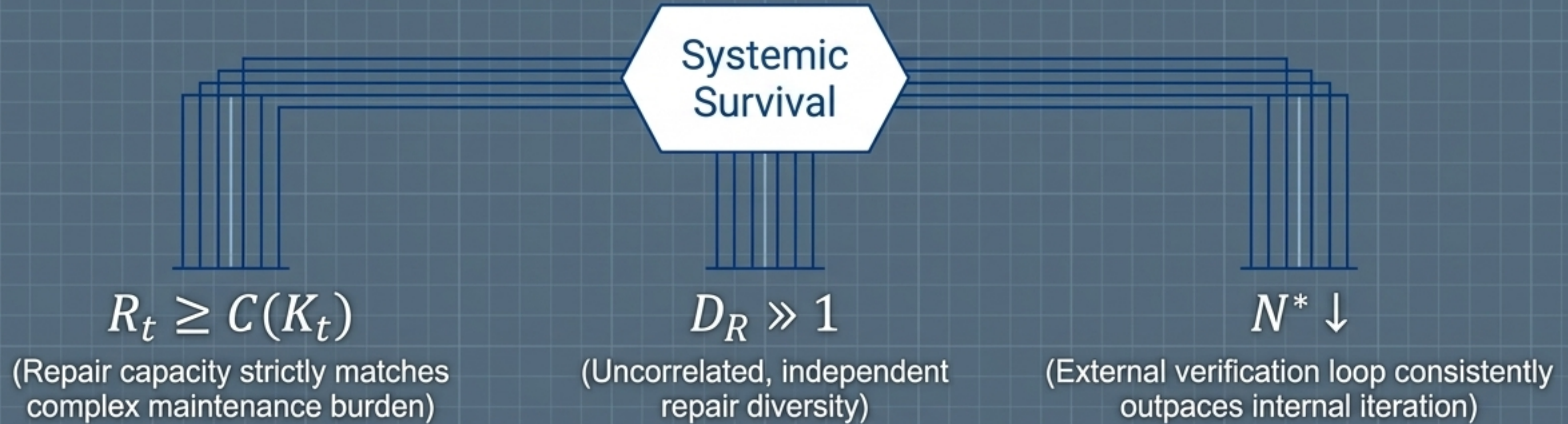


Civilizational institutions—courts, patent systems, archives, distributed markets—are mechanisms explicitly evolved to keep D_R high and to manufacture repair capacity (R_t) for the economic and legal recursion loop.

The Ultimate Synthesis: Evaluating AGI against historical precedents.

	Science	Civilization	Artificial General Intelligence (AGI)
Locus Count (n)	$n \gg 1$ (Global)	$n \gg 1$ (Global)	→ 1 (Monolithic / Centralized)
Iteration / Verification Ratio (N*)	Low (Decades)	Low (Centuries)	High (Computational timescales, $\Delta t \ll \tau$)
Effective Repair Diversity (D_R)	High	High	Low (→ 1, shared architectures and corpora)
Admissibility Status	Stable	Stable	Extreme Compression Risk

The Geometry of Recursive Systems



- The danger of AGI is not that it recurses. Recursion is an ancient, ubiquitous mechanism requiring no intelligence.
- The unique, existential danger lies in topological compression: attempting civilizational-scale capability accumulation inside a substrate with few independent loci, short iteration periods, and correlated, non-diverse repair mechanisms.
- Survival is not fundamentally a question of intelligence. It is a question of geometry.